Impact of Solar Activity on the Calculation of ToA Reflectance Values of Multispectral Images

Jonathan G. Müller, Alexander C. Vibrans, and Cibele Teixeira Pinto

***Abstract*— Solar activity explains a fraction of the variation in incident solar irradiance at the top of atmosphere (ToA). The radiometric calibration of multispectral sensors depends on the exoatmospheric spectral irradiance (ESUN) parameter that quantifies the incidence of energy on the ToA in spectral intervals. Since ESUN values are normalized, they neglect the variability caused by this phenomenon. Therefore, we investigated the impact of the variability of the ToA solar irradiance caused by solar cycles and activities on the computation of the ToA reflectance values of the Enhanced Thematic Mapper Plus (ETM+)/L7 sensor using daily ToA solar irradiance data from the Spectral Irradiance Monitor (SIM)/Solar Radiation and Climate Experiment (SORCE) instrument. The relative variability in the time series of ToA reflectance was computed during solar cycles number 23 (partially) and 24 (2003–2020). The results indicate that short-term solar activity can affect daily ToA reflectance up to 0.46%, 0.40%, 0.31%, 0.22%, 0.35%, and 2.37% in the blue, green, red, near-infrared (NIR), shortwave infrared (SWIR)-1, and SWIR-2 bands of the ETM+ sensor, respectively. However, when compared to the upper limits of the long-term uncertainties of ∼1.0% (visible (VIS)-NIR), ∼1.2% (SWIR-1), and ∼1.4% (SWIR-2), it appears that only the larger variations can be considered as due to the Sun's activity. The intrinsic long-term variations of the solar cycle are below these limits and may be influenced by the uncertainties of the instruments used. We conclude that solar activity is not negligible within radiometric calibration of multispectral sensors, but more accurate ToA solar irradiance data are needed to assess the impact of long-term phenomena.**

***Index Terms*— Mean exoatmospheric solar irradiance (ESUN), radiometric calibration, solar irradiance top of atmosphere (ToA), Solar Radiation and Climate Experiment (SORCE), Spectral Irradiance Monitor (SIM).**

I. Introduction

OBTAINING information through remote sensing techniques has led to important advances in the generation of data on atmospheric conditions, land cover, and oceans [1]. The use and application of this data has been and will continue to be essential in making important discoveries about climate change [1] and in generating predictor variables for climate data [2], [3], [4]. This study discusses issues that can contribute to improving the accuracy of top-of-atmosphere (ToA) reflectance data obtained by passive remote sensors.

The Sun is a gaseous sphere with differential rotation [5]. This rotation effect causes the twisting of its magnetic field lines, which, consequently, promote the occurrence of energetic events on the surface and in the solar atmosphere that provide the variability in the intensity of electromagnetic radiation (EMR) emitted [5], [6], [7].

Among these events, there is the occurrence of sunspots and explosions called solar flares and coronal mass ejections (CMEs). These three phenomena happen periodically on the surface and in the solar atmosphere, in periods of approximately 11 years on average, and are indicators of the Sun's activity [6], [7]. These periods are known as 11-year solar cycles (from here on, the solar cycles of 11 years will be called solar cycles or cycles of the Sun) and will be analyzed as long-term variations throughout this study. The isolated occurrence of solar activity events (spots, flares, and CMEs) also causes day interval fluctuations in the EMR emitted by the Sun, so they will be interpreted as short-term variations during the proposed analyses. The increase in brightness caused in the visible (VIS) and infrared spectra during the occurrence of flares and CMEs is called Faculae [7], [9]. Thus, the balance of electromagnetic energy emitted by the Sun during the period of a solar cycle is not constant. Its variability is due to a continuous "competition" between the fluctuations resulting from the isolated occurrence of each of the solar activity events [8], [10], [11].

Solar irradiance incident on the ToA is the fundamental physical principle of passive remote sensing techniques [12]. These techniques enable the collection of EMR data reflected and/or emitted by targets toward the field of view of a sensor. The recorded signals are discretized and converted into three different quantities called digital number (DN), radiance, and reflectance [12], [13]. Reflectance can be defined as a rate, obtained from the ratio between the energy reflected by the target in the direction of the sensor (radiance) and the energy incident on the period of passage of the sensor (irradiance) [12], [14].

In the current practice, to obtain the reflectance values of the targets, the ToA solar irradiance data are normalized by a constant parameter, called mean exoatmospheric solar irradiance (ESUN) [15]. Its function is to quantify the ToA solar irradiance in the spectral regions of operation of the sensor to enable the calculation of ToA reflectance in each of the

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bands. However, the implementation of this parameter ends up neglecting the variability caused by solar cycles and activities in this quantity and, consequently, in the characterization of the targets from the reflectance values.

In this context, the overall objective of this research is to investigate the possible effects of variations caused by solar cycles (long term) and solar activities (short term) on ToA solar irradiance for the computation of ToA reflectance values. The specific objectives are as follows.

1) Determine the amplitude of the variations caused by solar cycles and activities in the ToA solar irradiance in the spectral regions of operation of blue, green, red, NIR, SWIR-1, and SWIR-2 spectral bands of the Enhanced Thematic Mapper Plus (ETM+) sensor, onboard Landsat 7 (L7).

2) Evaluate the impacts, and the associated uncertainties, of the variations caused by the Sun's cycles and activities on the ToA solar irradiance for the calculation of the ToA reflectance values of the ETM+/L7 sensor. The underlying hypothesis of this study is that there are short- and long-term nonseasonal variations in the amount of ToA solar irradiance, caused by solar cycles and activities, which are not negligible for the computation of ToA reflectance from multispectral images.

II. Materials and Methods

A. Solar Radiation and Climate Experiment Data

The ToA solar spectral irradiance data used in this study were obtained by the Solar Radiation and Climate Experiment (SORCE) satellite mission. SORCE carried onboard four instruments that record daily readings of total and spectral ToA solar irradiance from April 2003 to February 2020 [8], [16]. This study used the Spectral Irradiance Monitor (SIM) database.

The SIM aboard SORCE acquired measurements of solar spectral irradiance in the near ultraviolet, VIS, near-infrared (NIR), and shortwave infrared (SWIR) ranges. The most updated data version of this instrument was used in this study—SORCE-SIM-TAV V02 (https://lasp.colorado.edu/home/sorce/data/).

The SIM/SORCE instrument is a spectroradiometer that covers the spectral range between 310 and 2400 nm and has a spectral resolution between 1 and 27 nm. The absolute accuracy of the data generated is 2%, its precision is <150 ppm (0.015%), and its long-term stability is 0.03% per year [16], [17], [18]. The signal-to-noise ratio (SNR) of the SIM/SORCE instrument is greater than 10^4 for spectral solar irradiance data below the 2000-nm wavelength and greater than 10^3 for spectral ranges greater than 2000 nm [19]. The irradiance data, obtained daily, are available for an average distance of one astronomical unit (1 AU) and the relative line-of-sight velocity is zero relative to the Sun [16], [17].

B. Landsat 7 ETM+ Sensor Data

The L7 was launched on April 15, 1999, containing onboard the ETM+ [20]. Among them, the main characteristics of the mission is its altitude of 705 km, inclination of 98.2°, and polar orbit, synchronous with the Sun. The imaged range is 185 km with a passage time between 10 and 12 pm (local time) and a revisit period of 16 days [20]. The ETM+ has a spatial resolution of 30 m (for the VIS, NIR, and SWIR bands) and a radiometric resolution of 8 bits [20]. Its degradation varies by −0.21% per year for the blue and SWIR-2 bands, −0.18% for SWIR-1, −0.13% in the NIR, −0.09 in the green, and −0.06% in the red [21]. It was an important tool for global land use and land cover monitoring.

To conduct the proposed study, we used spectral information from bands of blue (441–514 nm), green (519–601 nm), red (631–692), NIR (772–898 nm), SWIR-1 (1547–1749 nm), and SWIR-2 (2064–2345 nm) of the ETM+ sensor [20]. The values used as reference ESUN for their respective bands are: 1970 $\mathrm{Wm^{-2}\mu m^{-1}}$ for blue, 1842 $\mathrm{Wm^{-2}\mu m^{-1}}$ for green, 1547 $\mathrm{Wm^{-2}\mu m^{-1}}$ for red, 1044 $\mathrm{Wm^{-2}\mu m^{-1}}$ for NIR, 225.7 $\mathrm{Wm^{-2}\mu m^{-1}}$ for SWIR-1, and 82.06 $\mathrm{Wm^{-2}\mu m^{-1}}$ for SWIR-2. The spectral response functions of the ETM+ bands are available at https://Landsat.usgs.gov/spectral-characteristics-viewer.

C. Calculation of ToA Reflectance

The calculation of the ToA reflectance of a target imaged by a remote sensor can be defined according to the following equation [20], [22]:

$$\rho_0 = \frac{\pi L_0 d^2}{\mathrm{ESUN}\cos(\theta_z)} \tag{1}$$

where ρ_0 is the ToA reflectance of the imaged target (dimensionless), L_0 is the radiance recorded by the ETM+ sensor (in $\mathrm{Wm^{-2}\mu m^{-1}sr^{-1}}$), d is the Earth–Sun distance (in AU), and θ_Z is the solar zenith angle. The parameters d and θ_Z are responsible for the radiometric correction of the ESUN constant as a function of the seasonal variations of the ToA solar irradiance [14]. The radiance (L_0) can be related to its respective DN through the following equation [20]:

$$L_0 = \left(\frac{L_{\max} - L_{\min}}{\mathrm{DN}_{\max} - \mathrm{DN}_{\min}}\right) \cdot (\mathrm{DN} - \mathrm{DN}_{\min}) + L_{\min} \tag{2}$$

where $L_{\max}$ and $L_{\min}$ are the maximum and minimum radiance recorded in the respective ETM+ band, respectively; and $\mathrm{DN}_{\max}$ and $\mathrm{DN}_{\min}$ are the maximum and minimum DN in that band, respectively.

DN values are sensitive to the spacecraft's temperature, which varies according to the Earth–Sun distance, and consequently may be subject to seasonal disturbances in their values. Therefore, to prioritize the analysis of the nonseasonal impacts of solar cycles and activities when calculating ToA reflectance values, constant values were assumed for L_0, θ_Z, and d (DN = 125, $\theta_Z = 30°$, and $d = 1$) when calculating ToA reflectance values.

Under these conditions, two time series with daily ToA reflectance values were calculated for each band of ETM+. The first, called reference ToA reflectance, was obtained considering the constant ESUN parameter, according to the value suggested in the L7 handbook. The second, called calculated ToA reflectance, was determined considering the variable ESUN parameter, according to the daily values calculated for ESUN based on data from the SIM/SORCE instrument (the methodology for calculating ESUN values is presented in the supplementary materials).

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The analysis of relative variation of the daily data was performed in each of the time series obtained. Two approaches were made. The first one in relation to the reference solar spectra suggested in the L7 handbook for each band of ETM+ and the second one in function of the average values of the respective time series analyzed.

D. Calculation of Uncertainties

The uncertainties calculation for the daily values of the calculated ToA reflectance was performed by implementing the Monte Carlo method (MCM) [24], [25]. The mathematical model used to calculate the uncertainties is defined in (1). A normal pdf was used because the mathematical model adopted depends on only one variable, ESUN. The other parameters in (1) were fixed and their respective uncertainties were disregarded to isolate only the uncertainties arising from nonseasonal phenomena related to solar cycles and activities. The uncertainties obtained for the ESUN values are characterized as systematic uncertainties. The input parameters of the normal pdf are the calculated value for the ESUN (expected value) and its respective associated uncertainty (calculated through the MCM). The methodology, as well as the values obtained for the ESUN and ToA solar irradiance quantities, together with their respective uncertainties, is presented in the supplementary materials.

As the probability of spanning assumed in this study was two standard deviations (2σ), approximately $N \approx 220\,000$ iterations were performed on the distribution of each variable [25]. Based on these N distributions of the input variables (ESUN), a new distribution of N values was generated for the output variable (ToA reflectance). A confidence level of 95.45% was assumed during the calculations.

The uncertainty calculation was performed in two ways: 1) considering only the standard uncertainties of the input variables and 2) adding the uncertainties obtained in 1) with long-term stability of the SIM/SORCE sensor as follows [25]:

$$\sigma_{\text{FINAL}} = \sqrt{(\sigma_{\text{STANDARD}})^2 + (\sigma_{\text{STABILITY}} \cdot t)^2} \tag{3}$$

where σ_{FINAL} is the uncertainty value obtained in 2), σ_{STANDARD} is the uncertainty value obtained in 1), $\sigma_{\text{STABILITY}}$ is the long-term stability value of SIM/SORCE, and t is the day corresponding to the data being analyzed.

III. Results

The results indicate that solar cycles and activities impact the amount of solar energy incident on the ToA and the magnitude of this impact varies depending on the spectral range. It was found that the influence of solar activities can impact the calculation of ToA reflectance values at percentage levels higher than 0.4% (blue), 0.3% (green, red, and SWIR-1), 0.2% (NIR), and 2% (SWIR-2) (Fig. 1 and Table I).

The values obtained for the coefficient of variation (CV) (Table I) indicate the percentage of standard dispersion (1σ) of the data in each case and can therefore be used to quantify the influence of solar cycles on ToA solar irradiance values. For the blue, green, red, NIR, and SWIR-1 bands, the standard variability (1σ) is approximately 0.05% (or 0.10% considering 2σ), and in the SWIR-2 band, it is approximately 0.31%

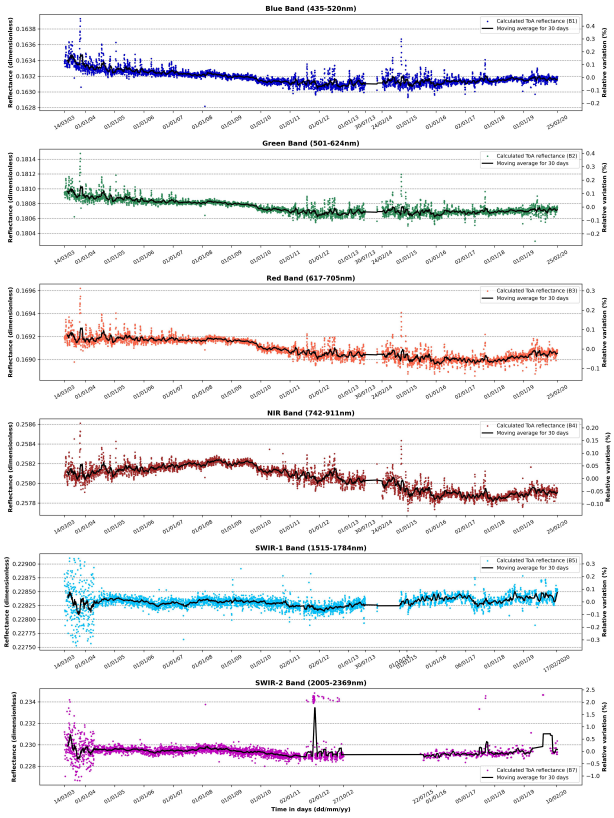

Fig. 1. Daily values calculated for the ToA reflectance in the spectral regions corresponding to the ETM+/L7 sensor bands considering DN = 125, solar zenith angle = 30°, and Earth–Sun distance = 1.0 AU.

(or 0.62% considering 2σ). The analysis of variation with respect to the reference ToA reflectance values indicated that the differences exceed 2.0%, 1.0%, 0.3%, 0.2%, 3.0%, and 4.0% in the blue, green, red, NIR, SWIR-1, and SWIR-2 bands, respectively (Table I), considering the possible noise accounted for in the SWIR-1 and SWIR-2 bands.

Since the effect caused by solar cycles (long term) is affected by the long-term stability of the sensors, the CV of the annual intervals of the ToA reflectance time series of the six analyzed ETM+ bands were calculated (Table II). Thus, it is possible to evaluate the impacts caused by short-term solar activity and reduce the effects caused by noisy data in the SWIR-1 and SWIR-2 bands. The results obtained for the annual CV (Table II) allow a comparison of the effects of solar activity in the different spectral regions observed and indicate that the greatest impacts occur in the blue, green, and SWIR regions.

The uncertainty calculation of the ToA reflectance values was performed in two conditions, omitting [Fig. 2(a)] and adding [Fig. 2(b)] the long-term stability uncertainty of the SIM/SORCE instrument in the used ESUN values. In the first condition, the results showed that the uncertainty obtained for the VIS and NIR bands is less than 0.2% in most of the analyzed period, while in SWIR-1 and SWIR-2, the uncertainties are approximately 0.6% and 0.9%, respectively, with exception of the periods corresponding to the years 2015, 2016, and 2017, which show a significant reduction in the values obtained, which are in line with the uncertainties available in the SORCE/SIM instrument database. In the second condition,

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TABLE I

SUMMARY STATISTICS OF THE CALCULATED TOA REFLECTANCE VALUES IN THE BANDS OF THE ETM+/L7 SENSOR CONSIDERING DN = 125, $\theta = 30°$, AND $D = 1.0$ AU

Band	Blue	Green	Red	NIR	SWIR1	SWIR2
N° days tested	5807	5806	5806	5801	5062	4308
Reference ToA reflectance [1]						
$\overline{\rho_R}$	0.1664	0.1825	0.1690	0.2579	0.2355	0.2252
$\overline{\Delta\rho_R}$	-1.9313	-0.9399	0.0341	0.0459	-3.0440	1.8785
$\min\Delta\rho_R$	-2.1515	-1.1922	-0.1051	-0.0813	-3.3800	0.6706
$\max\Delta\rho_R$	-1.4824	-0.5450	0.3469	0.2641	-2.7095	4.2968
Calculated ToA reflectance [2]						
$\overline{\rho_0}$	0.1632	0.1808	0.1691	0.2580	0.2283	0.2294
$\min\rho_0$	0.1628	0.1803	0.1687	0.2577	0.2275	0.2267
$\max\rho_0$	0.1639	0.1815	0.1696	0.2586	0.2291	0.2348
σ_{ρ_0}	0.0001	0.0001	0.0001	0.0001	0.0001	0.0007
CV_{ρ_0}	0.0561	0.0523	0.0499	0.0510	0.0530	0.3136
$\min\Delta\rho_0$	-0.2206	-0.2600	-0.1360	-0.1279	-0.3460	-1.1858
$\max\Delta\rho_0$	0.4596	0.3983	0.3134	0.2170	0.3460	2.3715

Where $\overline{\rho_R}$ is the mean of the reference ToA reflectance values (dimensionless); $\overline{\Delta\rho_R}$, $\min\Delta\rho_R$ and $\max\Delta\rho_R$ are the mean, minimum and maximum percentage variations between the calculated ToA reflectance values and reference ToA reflectance (%); $\overline{\rho_0}$, σ_{ρ_0}, CV_{ρ_0}, $\min\rho_0$ and $\max\rho_0$ the mean, standard deviation (dimensionless), coefficient of variation, minimum value and maximum value (%) of the calculated ToA reflectance values; $\min\Delta\rho_0$ and $\max\Delta\rho_0$ the minimum and maximum percentage variations of the calculated ToA reflectance values relative to the mean value of the respective obtained time series (dimensionless).

[1]the ToA reflectance values obtained from the reference ESUN parameters.

[2]ToA reflectance values calculated from SIM/SORCE solar spectra.

TABLE II

ANNUAL COEFFICIENTS OF VARIATION OF THE CALCULATED TOA REFLECTANCE VALUES FOR THE BANDS OF THE ETM+/L7

Year	CV_{BLUE}	CV_{GREEN}	CV_{RED}	CV_{NIR}	CV_{SWIR1}	CV_{SWIR2}
2003	0.0672	0.0561	0.0459	0.0388	0.1562	0.57198
2004	0.0416	0.0341	0.0275	0.0237	0.0821	0.22178
2005	0.0329	0.0275	0.0220	0.0192	0.0226	0.08567
2006	0.0210	0.0170	0.0141	0.0131	0.0255	0.09320
2007	0.0155	0.0112	0.0091	0.0094	0.0283	0.09135
2008	0.0196	0.0121	0.0095	0.0109	0.0235	0.13920
2009	0.0157	0.0139	0.0130	0.0126	0.0307	0.10529
2010	0.0168	0.0139	0.0118	0.0135	0.0335	0.11161
2011	0.0354	0.0292	0.0246	0.0221	0.0389	0.74243
2012	0.0371	0.0307	0.0257	0.0249	0.0276	0.49618
2013	0.0311	0.0252	0.0199	0.0168	0.0308	---
2014	0.0552	0.0451	0.0372	0.0330	0.0325	---
2015	0.0371	0.0319	0.0283	0.0244	0.0333	0.08597
2016	0.0240	0.0213	0.0198	0.0149	0.0270	0.08963
2017	0.0241	0.0210	0.0190	0.0188	0.0400	0.32960
2018	0.0135	0.0133	0.0165	0.0125	0.0395	0.11759
2019	0.0206	0.0263	0.0237	0.0165	0.0425	0.47927

the results indicate a gradual increase in uncertainties, due to the long-term stability considered, starting at the values reported in the first condition and reaching maximum values at the end of the analyzed period. The maximum uncertainties obtained in the second condition are approximately 1.0% for the VIS and NIR bands and 1.2% and 1.4% for the SWIR-1 and SWIR-2 bands, respectively, with exception of the lows of the years 2015, 2016, and 2017 in the SWIR-2 band.

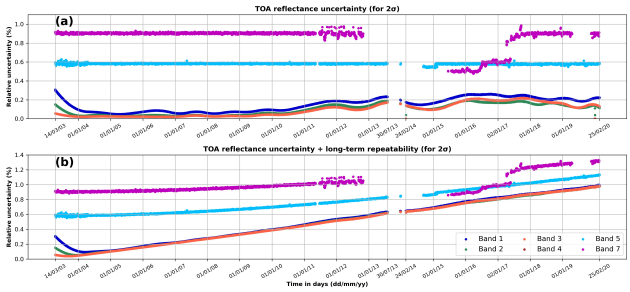

Fig. 2. Relative uncertainties of the calculated ToA reflectance values for (a) spectral bands of the ETM+/L7 sensor neglecting and considering (b) long-term stability of SIM/SORCE.

The overlapping of the relative variations of the calculated ToA reflectance values with the respective uncertainties obtained indicates that only the highest intensity oscillations of cycle 23 can be considered as influenced by solar activity. The annual CVs also indicate that only the periods of maximum solar activity have quantified variations above the respective uncertainty levels. On the other hand, the variations of lower proportion, generally consistent with the periods of minimum solar activity, may be influenced by the uncertainty of the instruments used.

IV. DISCUSSION

Considering the time series obtained (Fig. 1), it is possible to identify the variations that occurred during solar cycles 23 and 24. Among the main aspects observed, the period of minimum solar activity between the years 2008 and 2009 stands out (signaling the end of cycle 23 and beginning of cycle 24), characterized by the lowest dispersion of data; and the periods of maximum solar activity of cycle 23, in the second half of the year 2003, and of cycle 24, in the year 2014, identified by the greatest dispersion of the calculated values. The largest impacts observed occurred during periods of maximum solar activity, characterized by the passage of large groups of sunspots and faculae in the field of view of the sensors, which slightly modify the solar brightness in these spectral bands during these time intervals (Fig. 1).

Previous studies have reported that differences in the quantification of the solar spectrum can compromise the retrieval of information obtained from remote sensing products in certain applications [15], [22], [23]. Reference [15] studied the effect of using six different solar spectra on the computation of ESUN values, ToA reflectance, and vegetation indices for two hyperspectral sensors. The results indicated significantly large differences in the calculated ESUN values and ToA reflectance corresponding to the SWIR bands. The largest observed differences in ToA reflectance values in SWIR were on the order of 0.01–0.12 for bright targets. The atmosphere resistant vegetation index (ARVI) vegetation index was shown to be more susceptible than the normalized difference vegetation index (NDVI) to variations in the solar spectrum. The authors indicate that the variation observed in the analyzed magnitudes may occur as a function of solar activity and the uncertainties of the measurement and calibration processes.

This article is distinguished by the proposal of analyzing separately the impacts of the cycles and activities of the Sun on the variation of solar spectra and measuring the

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propagation of these impacts to the ToA reflectance products. The results agree with the variability profile identified in the work of [15], however, with lower variation amplitudes and percentage difference.

In addition, this study analyzed the daily temporal variability of solar spectra to develop subsidies for the radiometric calibration of sensors that do not have integrated solar diffuser panel systems for the recording and calibration of the spectral solar irradiance indices used, as is the case of OLI/L8. The results obtained are aligned with the results presented by Li et al. [22] and point out that the magnitude of temporal variations in solar spectra is like the variability observed in ToA reflectance values. This aspect can be relevant for the improvement of the radiometric calibration process, because from the variations in the solar spectrum, one can estimate a calibration coefficient to correct such effect on the respective ToA reflectance values.

In recent years, there has been a growing need to establish instruments with traceability to the International System of Units (SI) in orbit [26]. Such instruments are essential to promote strategies that improve the quality of in-orbit calibration of sensors coupled to satellite missions and, consequently, provide more accurate and robust data on climate change and natural resource monitoring. These instruments are called SI-Trackable Satellites (SITSats) and can act as a reference in the intercalibration of sensors and generate more reliable long-term data [26]. This study can contribute to the development of future SITSats about the importance of considering nonseasonal variations in ToA solar irradiance caused by solar cycles and activities, as well as subsidizing the standardization of absolute values for daily solar spectra.

Li et al. [22] investigated the variation in recovered surface reflectance values from a multispectral sensor using different solar spectra and for different land uses. Their results indicated that a small variability in the solar spectrum influences the computation of ToA reflectance and can generate larger impacts on the atmospheric correction process for retrieval of surface reflectance values. The largest impacts identified by the authors were obtained for the band corresponding to the blue spectral region (VIS) and for low reflectance targets (dark targets), such as aquatic applications. In the NIR and SWIR bands, the impacts presented were smaller. The authors also discuss the importance of using more continuous solar spectra to improve the calibration of the solar constant used in remote sensing techniques.

The results of this study also indicate that the percentage variation of ESUN values from the ETM+ reference solar spectrum is greater in the SWIR spectral region, as reported by Bhatt et al. [23]. The magnitude of the differences identified here for the VIS and SWIR regions are as high as 2.1% and 4.1%, respectively. The work of [23] had the objective of investigating the impacts of using different solar spectra on intercalibration between two multispectral remote sensors and retrieving environmental information through multiband reflectance measurements. The results indicated that ESUN values obtained from eight different solar spectra can differ by more than 2% in VIS and 10% in SWIR from a reference spectrum (Thuillier 2003). According to the authors, these differences can be intensified when retrieving environmental information from the use of multiple bands from one sensor, such as the NDVI, the normalized difference snow index (NDSI), or the ratio between bands.

Overall, the goal of the three listed studies [15], [22], [23] was to quantify the impact of using different solar spectra (since there are several suggested in the scientific literature) on remote sensing products. The authors reach a common conclusion that more accurate values of the ESUN parameter can provide more accurate results for reflectance. In this context, this article differs by investigating and quantifying the influence of solar cycles and activities as a source of uncertainty for the computation of the ESUN parameter and, consequently, for the radiometric calibration of the ToA reflectance values.

The use of reflectance obtained by multispectral sensors in the VIS, NIR, and SWIR regions is essential resources for obtaining large-scale information on environmental variables and for monitoring land use and occupation [1]. This information, obtained through instruments that were not originally developed for this purpose, can be implemented in mathematical models as predictor variables for long-term climate data. For example, Zakeri and Mariethoz [4] used satellite data to generate a 20-year time series with daily images of synthetic snow cover in Switzerland; Guo et al. [3] used long-term NDVI (1982–2015) to analyze spatial–temporal patterns of vegetation activity and thus infer about temperature, precipitation, soil moisture, and anthropogenic activity, and Hamilton et al. [2] investigated climate change by monitoring seaweed populations based on a 35-year historical series of Landsat images.

The set of results obtained in this study is possible to infer that certain sets of sunspots and faculae during periods of maximum activity of the Sun influence the ToA reflectance products. However, more accurate solar spectral irradiance data is needed, at least lower than the coefficients of variation of the series analyzed.

V. Conclusion

Based on our results, it is possible to assert that there is no evidence to refute the hypothesis of this study. The main conclusions are given as follows.

1) The variations caused by solar activities during the cycles of the Sun are not negligible in the radiometric correction process of multispectral remote sensors and, therefore, should be considered when generating ToA reflectance data.

2) Adjusted coefficients should be established for the radiometric correction of ToA reflectance from missions without integrated solar diffuser panels for direct ToA solar irradiance measurements. This correction of remote sensing products, including those from missions that are inoperative, could have an impact on the generation of long-term climate data from missions that were not originally designed for this purpose.

3) There is a need to develop and use more accurate ToA solar spectral irradiance measurement instruments than SIM/SORCE, in order to estimate more precisely the contribution of solar cycles to uncertainties of multispectral ToA reflectance products.

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